

# 1 Soundness of the thresholded verifier

A posterior over theories can serve as a proof checker: a prover proposes a sentence, and the verifier accepts it when almost all posterior mass sits on theories that derive it. Is this sound against provers that adapt to it (the brief’s H3)? With fixed weights and well-specified data, yes: with probability at least  $1 - \delta/\pi(C_d^*)$  no invalid sentence is ever accepted (Theorem 1.2). With the default Dirichlet weights, on average over the weight prior, or at each weight vector with a threshold that shrinks with  $n$ , but not with a constant threshold (Theorems 1.10 and 1.11 and Example 1.12). Under misspecification there is no guarantee; in an example a waiting prover wins with probability 1 (Proposition 1.5). The limit  $\delta \rightarrow 0$  is the cautious verifier of AS, read on theorems (Proposition 1.6). In Hänni’s triple “true / false / independent”, Bel is a belief function and Pl its plausibility (Section 1.6). Sources: model §§4, 7; experiments Prop X8, E2, E4; universal §§6.2, 11. Proofs are in Appendix A.

## 1.1 Protocol

**Definition 1.1** (Protocol; thresholded verifier). <sup>model §4</sup> Fix a countable class  $\mathcal{H}$  with prior  $\pi$ , a truth  $T^* \in \mathcal{H}$  and a derivability bound  $d \leq \infty$ . Data  $X_1, X_2, \dots$  form an infinite sequence. In each round  $t = 1, 2, \dots$  either the next datum arrives or a *prover* submits a query  $q_t$ ; the prover chooses which. Let  $n(t)$  be the number of data that have arrived by round  $t$ . The verifier  $V_{\delta,d}$  answers **ACCEPT** iff  $\pi_t(\{T : q_t \notin \text{Th}_d(T)\}) \leq \delta$ . Otherwise it may **ESCALATE** to a deterministic oracle that says whether  $q_t \in \text{Th}_d(T^*)$ ; the  $j$ -th answer becomes the constraint  $c_j(T) := 1[q \in \text{Th}_d(T) \Leftrightarrow q \in \text{Th}_d(T^*)]$ . The posterior at round  $t$  is  $\pi_t(T) \propto \pi(T) \prod_{i \leq n(t)} P_T(X_i) \prod_j c_j(T)$  with fixed weights; with Dirichlet weights the marginal  $P_T^{\text{Dir}}(X_1, \dots, X_{n(t)})$  replaces the product over data. The verifier is *sound* on a run if it never accepts a sentence outside  $\text{Th}_d(T^*)$ . Finally  $C_d^* := \{T : P_T = P_{T^*}, \text{Th}_d(T) = \text{Th}_d(T^*)\}$  and  $W_d^* := \pi(C_d^*) \geq \pi(T^*)$ .

The prover may be randomised, adaptive and computationally unbounded, may know  $T^*$ , the verifier and the whole future data sequence, and chooses when data arrive. Soundness is relative to  $\text{Th}_d(T^*)$ , not to truth. The unit of  $d$  is symbol size (??); the experiments use chain steps ( $\vdash_J$ ), and pa’s example in Remark 1.14 uses  $d = 0$  (being an instance). This is the protocol of AS Def 2.6 with  $V_{\delta,d}$  in place of the version-space verifier.

## 1.2 Fixed weights

**Theorem 1.2** (Time-uniform soundness, fixed weights). *[proved; computed]<sup>model Thm 4.1</sup> Assume setting  $W$  (??): every  $T \in \mathcal{H}$  carries a fixed law  $P_T$  (fixed mixture weights), the data are i.i.d. from  $P_{T^*}$  with  $\pi(T^*) > 0$ , the prior does not depend on  $n$ , and the oracle is truthful. If  $\delta \leq W_d^* \delta'$ , there is an event  $G$  with  $P(G) \geq 1 - \delta'$ , depending only on the data sequence, on which no prover ever gets a sentence outside  $\text{Th}_d(T^*)$  accepted, in any round.*

*Proof idea.* IL Thm 4.15(b) in the class form of IL Thm 10.9, with  $C_d^*$  as the target. Members of  $C_d^*$  share  $T^*$ ’s likelihood and pass every constraint, so  $\pi_t(C_d^*) = W_d^*/Z_t$  with  $Z_t \leq M(X^n)/P_{T^*}(X^n)$ ,  $n = n(t)$ ,  $M$  the Bayes mixture. That ratio is a nonnegative supermartingale started at 1, so by Ville’s inequality (IL Lemma C.4; Ville, 1939) it ever reaches  $1/\delta'$  with probability at most  $\delta'$ . Elsewhere  $\pi_t(C_d^*) > \delta$  in every round, and no member of  $C_d^*$  derives a sentence outside  $\text{Th}_d(T^*)$  (Appendix A.1).  $\square$

One event covers all queries and rounds, so no union bound over the prover’s attempts is needed. The brief’s H3 is confirmed for fixed weights, with the constant improved from  $\pi(T^*)$  to

$W_d^*$ . The guarantee is per target: under  $\pi_\lambda$  it covers the  $T^*$  with  $\lambda\ell(T^*) \leq \log_2(\delta'/\delta)$ , so for long axiom systems it is weak. IL's escalation bound (IL Thm 4.16) transfers with  $C_d^*$  as the target. Read at the single sentence  $\forall x\varphi$ , with the class of members of  $C^*$  that agree with  $T^*$  on it, the theorem gives ???. In simulations the frequency of invalid acceptance stayed 2–5 times below the bound (Appendix A.1).

*Remark 1.3* (The constant is tight). [proved]<sup>model §4.1 (referee m13)</sup> IL Prop C.5 shows that  $\delta \leq W^*\delta'$  cannot be improved, with  $R^* = \{a, b\}$  against  $R' = \{a, c\}$ , which generates only  $a$  and proves  $c$ . That pair is not in the template models ( $\text{supp } P_{T'} \subseteq \text{supp } P_{T^*}$  implies  $\text{Th}_d(T') \subseteq \text{Th}_d(T^*)$  under L0, and under L1 with parameters admissible). But  $T' = \{a, c\}$  with a small fixed weight on  $c$  attains IL's acceptance probability  $(1 - u)^{t^*}$  exactly, and it tends to  $\delta'$  as  $u, W^* \rightarrow 0$  (Appendix A.1).

*Remark 1.4* (Hypotheses). [proved; refuted (two old remarks)]<sup>model §4.1 (referee m8)</sup> What matters: (i) well-specification of the *data law*, not only of the theorem set (Proposition 1.5; selected data in Remark 1.13(A)); (ii) fixed weights, or Section 1.4; (iii) a proper prior that does not depend on  $n$  (a penalty  $\lambda_n$  growing with  $n$  breaks the supermartingale step; no replacement is known); (iv) truthful constraints. A finite truncation  $\mathcal{H}_N \ni T^*$  keeps the theorem with  $W_d^*$  replaced by  $\pi(C_d^* \cap \mathcal{H}_N)/\pi(\mathcal{H}_N) \geq \pi(T^*)$ ; so does a computable verifier that computes the non-deriving mass to within  $\varepsilon$  (??(a)) and accepts iff the upper end is  $\leq \delta$ . **Refuted** (model track, first version): “ $W^*$  only grows under truncation” (the ratio above can fall to 0.09 against  $\pi(C_d^*) = 0.90$ ) and “L2 likelihoods are finite sums” (not with full-support  $\mathcal{Q}$ ); see Appendix A.1.

**Proposition 1.5** (Misspecification: a waiting prover wins). [proved; computed]<sup>model Prop 4.2</sup> *In the setting of ?? (human axioms  $\mathcal{Q}$ ; data  $t = 0$  for closed  $\{0, +\}$ -terms  $t$  from a law with infinite support; a finite class of sound theories, possibly with  $\mathcal{Q}$ , plus  $T_{\text{esc}} = \{z = 0\}$ ; L0), let the prover wait until  $\pi_t(\{T : T \not\models S0 = 0\}) \leq \delta$  and then submit  $S0 = 0$ . For every  $\delta > 0$ , with probability 1 the verifier  $V_{\delta, \infty}$  accepts  $S0 = 0$  at some finite time, whether or not it escalates; yet  $\mathcal{Q} \vdash \neg S0 = 0$ .*

*Proof.* By ??, almost surely  $\pi_n(T_{\text{esc}}) = 1$  for all large  $n$ , and  $T_{\text{esc}} \vdash S0 = 0$ . The prover's only query is made when it is accepted.  $\square$

Computed: accepted in 2000/2000 runs (median 17 data); against an escalating verifier the waiting prover won 400/400, a prover asking in every round 11/400 (Appendix A.1). Misspecification can also make the verifier incomplete instead: in ?? it never accepts  $\forall x(0 + x = x)$ . Which happens depends on whether the best-fitting generator is weaker than  $T^*$  or incomparable with it.

### 1.3 The cautious limit and completeness

**Proposition 1.6** (The limit  $\delta \rightarrow 0$ ). [proved]<sup>model Prop 4.3</sup> *Suppose each  $P_T$  has a fixed support  $\text{supp}_T$  (fixed weights; or Dirichlet weights under L0, where  $P_T^{\text{Dir}}(D) > 0$  iff  $D \subseteq \bigcup \text{inst}(T)$ ).*

- (a)  $\pi_n(T) > 0$  iff  $\pi(T) > 0$  and  $D_n \subseteq \text{supp}_T$ . So  $V_{0,d}$  accepts exactly  $\bigcap \{\text{Th}_d(T) : \pi(T) > 0, D_n \subseteq \text{supp}_T\}$ .
- (b)  $V_{0,d}$  is sound for every  $T^*$  with  $\pi(T^*) > 0$  and  $D_n \subseteq \text{supp}_{T^*}$ , deterministically and without well-specification (IL Thm 4.14).  $V_{\delta,d}$  accepts at least as much.
- (c) Under L0 with full-support  $\mathcal{Q}$ , if  $\pi$  charges every finite theory,  $V_{0,d}$  accepts exactly  $\text{Th}_d(D_n)$  (AS Prop 6.4, read on theorems).

- (d) Under  $\mathbf{L0}$ , if  $\text{supp } \pi = H_k(\text{DT}^\circ)$ , the acceptance set of  $V_{0,d}$  contains  $\text{Th}_d(\text{Acc}_k(D_n))$ , where  $\text{Acc}_k$  is the cautious  $k$ -union verifier of AS Def 2.6. In general classes the inclusion can be strict; for  $H_k(\text{DT}^\circ)$  itself this is open.

**Proposition 1.7** (Completeness without anchors). *[proved]<sup>model Prop 4.4</sup> Assume setting  $W$  with  $\mathbf{L0}$  (full-support  $\mathcal{Q}$ , any  $d$ ) or with  $\mathbf{L1}$  ( $d = \infty$ , parameters admissible). For every  $s \in \text{Th}_d(T^*)$  and every  $\delta > 0$ , almost surely  $V_{\delta,d}$  accepts  $s$  at all sufficiently large  $n$ . With Dirichlet weights under  $\mathbf{L0}$  the same holds for Lebesgue-almost every  $w^*$ .*

*Proof.* By ?? the non-deriving mass of  $s$  tends to  $\pi(C^* \cap \{T : s \notin \text{Th}_d(T)\})/\pi(C^*)$ , which is 0 because under these likelihoods every member of  $C^*$  has the theorems of  $T^*$ . With Dirichlet weights use ??(b).  $\square$

The cautious verifier becomes exact only once the data contain an anchor (AS Lemma 2.8), and without a bound on the number of templates it accepts only the data (AS Prop 6.4). The Bayesian verifier needs neither, because the likelihood moves mass away from memorisers and splits; it gives up deterministic soundness. Table 1 (Appendix A.2) sets the two verifiers side by side. The time to accept  $s$  is set by the competitors that miss it: a split that omits a root of probability  $p$  gains a factor  $(1-p)^{-n}$  until the first datum with that root.

## 1.4 Dirichlet weights

*Remark 1.8* (Vacuity for Dirichlet weights). *[proved; refuted (earlier use)]<sup>model Rem 4.5; experiments §0 (M3)</sup> Read Theorem 1.2 with parameter  $(T, w)$  and prior  $\pi(T)\text{Dir}_T(dw)$ . Its class  $\{(T, w) : P_{T,w} = P_{T^*,w^*}, \text{Th}_d(T) = \text{Th}_d(T^*)\}$  has prior mass 0 if  $|T^*| \geq 2$ , the component laws of  $T^*$  are linearly independent, and  $w^*$  lies outside a countable set; then  $\delta \leq W^*\delta'$  forces  $\delta = 0$ . The model track's first version applied the theorem to the Dirichlet model, and the experiments' first summary promised soundness at a constant threshold for any well-specified data; both are **refuted** (Example 1.12).*

**Lemma 1.9** (Regret). *[(a), (b), (d) proved; (c) known; computed]<sup>model Lemma 4.6; experiments Prop X8(c)</sup> Let  $T$  have  $K$  templates with laws  $q_\tau$  (under  $\mathbf{L0}$ ,  $q_\tau = \mathcal{Q}_\tau$ ),  $P_{T,w} := \sum_\tau w_\tau q_\tau$ , weights  $\text{Dir}(\alpha)$ ,  $A := \sum_\tau \alpha_\tau$ , and*

$$R_\alpha(n, K) := \min_{c \in \mathbb{N}^K, \sum_\tau c_\tau = n} \frac{\mathbb{E}_{\text{Dir}(\alpha)}[\prod_\tau w_\tau^{c_\tau}]}{\prod_\tau (c_\tau/n)^{c_\tau}} \quad (0^0 := 1).$$

(a)  $P_T^{\text{Dir}}(x^n) \geq R_\alpha(n, K) P_{T,w}(x^n)$  for every  $w$  in the closed simplex and every  $x^n$ .

(b) If  $\alpha_\tau \leq 1$  for every  $\tau$ , then  $R_\alpha(n, K) \geq \frac{\Gamma(A)}{e(K-1)! \prod_\tau \Gamma(\alpha_\tau)} (n+1)^{-(K-1)}$ .

(c) For  $\alpha_\tau \equiv \frac{1}{2}$ :  $-\ln R_\alpha(n, K) = \frac{K-1}{2} \ln n + O(1)$  (Krichevsky and Trofimov, 1981).

(d) For  $\alpha_\tau \equiv \alpha$ :  $R_\alpha(n, K) \leq \frac{\Gamma(K\alpha)\Gamma(\alpha+n)}{\Gamma(\alpha)\Gamma(K\alpha+n)} \sim \frac{\Gamma(K\alpha)}{\Gamma(\alpha)} n^{-(K-1)\alpha}$ ; so at  $\alpha = 1$  the exponent  $K-1$  in (b) is tight.

(a) compares the two sides term by term over labellings of the data; (b) integrates over a copy of the simplex shrunk by  $1/(n+1)$  towards  $c/n$ ; (d) takes  $c = (n, 0, \dots, 0)$  (Appendix A.3).

**Theorem 1.10** (Soundness on average over  $w^*$ ). *[proved]<sup>model Thm 4.7; experiments Prop X8(a), (b)</sup> Let  $\mathcal{H}$  be countable with prior  $\pi$  and Dirichlet weights, each  $T$  with its marginal law  $P_T^{\text{Dir}}$  on data sequences (under L0, L1 or L2), and truthful constraints. Let the data come from  $P_{T^*}^{\text{Dir}}$ : first  $w^* \sim \text{Dir}_{T^*}$ , then i.i.d. from  $P_{T^*, w^*}$ . Put  $C_d^{\text{Dir}} := \{T : P_T^{\text{Dir}} = P_{T^*}^{\text{Dir}} \text{ on } \mathcal{S}^n \text{ for all } n, \text{Th}_d(T) = \text{Th}_d(T^*)\} \ni T^*$ . If  $\delta \leq \pi(C_d^{\text{Dir}}) \delta'$ , there is a data event  $G$  with  $P_{T^*}^{\text{Dir}}(G) \geq 1 - \delta'$  on which no prover ever gets a sentence outside  $\text{Th}_d(T^*)$  accepted. Hence  $\text{Dir}_{T^*}\{w^* : P_{T^*, w^*}^\infty(G^c) > \delta'/\gamma\} \leq \gamma$  for every  $\gamma \in (0, 1]$ . Pool version (experiments): with the experiments' prior weights  $2^{-\text{bits}(T)}$  and  $\vdash_J$  for  $\text{Th}_d$ , the verifier may use the posterior restricted to any pool  $R_n \ni T^*$ , even one chosen from the data, and the conclusion holds for  $\delta \leq 2^{-\text{bits}(T^*)} \delta'$ ; for a fixed pool  $H$  plus all memorisers,  $2^{-\text{bits}(T^*)}$  may be divided by  $\sum_{T \in H} 2^{-\text{bits}(T)} + 1$ .*

**Theorem 1.11** (Soundness at a fixed  $w^*$ ). *[proved]<sup>model Thm 4.8; experiments Prop X8(c)</sup> As in Theorem 1.10, but with data i.i.d. from  $P_{T^*, w^*}$  for a fixed  $w^*$  in the closed simplex, a likelihood linear in  $w$  (L0; also the experiments' chain  $\text{Ch}_J$ ),  $K := |T^*|$ , and a verifier that accepts at round  $t$  iff the non-deriving mass is at most  $\delta_{n(t)}$ . If  $\delta_n \leq \pi(C_d^{\text{Dir}}) \delta' R_\alpha(n, K)$  for every  $n$ , then with probability at least  $1 - \delta'$  no prover ever gets a sentence outside  $\text{Th}_d(T^*)$  accepted. For  $\alpha \leq 1$ , Lemma 1.9(b) gives an explicit  $\delta_n \propto (n+1)^{-(K-1)}$ ; with  $\alpha = \frac{1}{2}$  and the exact  $R_\alpha$ ,  $\delta_n \asymp n^{-(K-1)/2}$ . To cover every target with at most  $k$  templates, use  $\min_{K \leq k} R_\alpha(n, K)$ .*

*Proof idea for both.* For Theorem 1.10, run the proof of Theorem 1.2 under the sequence law  $P_{T^*}^{\text{Dir}}$ , then apply Markov's inequality in  $w^*$ . For Theorem 1.11, apply Ville's inequality to  $M/P_{T^*, w^*}$  and pass to the marginal  $P_{T^*}^{\text{Dir}}$ , which all of  $C_d^{\text{Dir}}$  shares, by Lemma 1.9(a) (Appendix A.3).  $\square$

Theorem 1.11 is time-uniform, but its threshold falls with  $n$  and slows acceptance; whether completeness (Proposition 1.7) survives is open. Theorem 1.10 covers a fixed  $w^*$  only outside a set of small prior mass. The shrinking threshold is needed:

**Example 1.12** (Constant threshold). *[computed; mechanism proof sketch]<sup>model Ex 4.9; experiments Prop X8</sup> L0; root law of  $\mathcal{Q}$   $(p_0, p_S, p_+, p_-) = (0.5, 0.5 - \varepsilon, \varepsilon/2, \varepsilon/2)$ ; truth  $T^* = \{0+0=0, (Sz)+0=Sz\}$  with  $\text{Dir}(\frac{1}{2}, \frac{1}{2})$  weights, data from  $P_{T^*, w^*}$  with the interior  $w^* = (p_0, p_S)/(1 - \varepsilon)$ . The competitor  $T' = \{z+0=z\}$  proves  $q := (0 \cdot 0) + 0 = 0 \cdot 0$ , which  $T^*$  does not. Prior  $\frac{1}{2}$  each; the verifier accepts  $q$  iff  $\pi_n(T^*) \leq \delta = 0.01$ , so Theorem 1.10's bound is 0.02. Two implementations checked every  $n \leq 2 \cdot 10^6$  with 300 runs per cell, a third a geometric grid of  $n$  (Table 2). The acceptance probability was 0.977–0.997 ( $\varepsilon = 10^{-5}$ ) and 1.000 ( $\varepsilon = 10^{-6}$ ) at the fixed  $w^*$ ; 0.003–0.010 with  $w^*$  drawn from the prior; and 0/300 at the fixed  $w^*$  with the threshold of Theorem 1.11.*

The mechanism: on the data's support  $P_{T'} = (1 - \varepsilon)P_{T^*, w^*}$ , and  $\ln P_{T'}(D) - \ln P_{T^*}^{\text{Dir}}(D)$  is  $n \ln(1 - \varepsilon)$  plus the regret  $\frac{1}{2} \ln n + O(1)$  of Lemma 1.9(c) plus an  $O_P(1)$  term. While  $\varepsilon n \ll 1$  it grows like  $\frac{1}{2} \ln n$  and crosses  $\ln 99$ , near  $n \approx 8000$  (Appendix A.3). With fixed weights the ratio is  $(1 - \varepsilon)^n \leq 1$  and  $q$  is never accepted.

## 1.5 What the experiments show

**Remark 1.13** (E4: adaptive provers in the code). *[computed]<sup>experiments E4</sup> Verifier on  $\vdash_1$ ,  $\text{Dir}(\frac{1}{2})$  weights, one-component  $T^*$ , so Theorem 1.10 applies (??). (A) The tight construction of IL Prop C.5 with a Dirichlet decoy  $T' = \{\varphi(0), 0 = S0\}$ : win probability exactly  $(1 - u)^{t^*}$ , 9–46 times below  $\delta'$  in five settings; the full pipeline agrees with the closed-form event on 400/400 streams in the two settings checked. With *selected* data (only  $\varphi(0)$  shown, each a theorem of  $T^*$ ) the prover wins 20/20 in every setting. (B) Well specified ( $\varphi = x+0 = x$ ,  $T^* = H_{\text{sch}}$ , bound 0.05,*

every  $n \leq 256$ ): 0/100 wins under L0 and L1; the largest mass on theories deriving an invalid query is 0.45 and 0.53, always on the true but underivable  $\forall x(x + 0 = x)$ , at  $n \leq 6$ . (C1) 10% false near misses  $t + 0 = St$ , with the mistake schema in the pool: 100/100 wins, first acceptance at the first mistaken datum in 90/100 streams. (C2) Heavy-tailed terms: 0/100. (C3) Selection with the decoy and a clean memoriser in the pool: 0/100.

In (C1) the verifier is sound for the data’s actual generator (a two-schema theory with fixed weights, covered by Theorem 1.11) and unsound for the intended  $T^*$ , as the brief’s H2 anticipated. Comparing (C3) with (A): under misspecification soundness depends on what else is in the pool, and no theorem covers this.

*Remark 1.14* (Early unsound lumps). [computed]<sup>experiments E2; pa §5.4</sup> E2 (L0,  $T^* = Q1..Q7 + T_{\text{Ind}}$  at fixed weights, causal pools with  $\text{SeenQ}(D_n)$ , 25 seeds, verifier on citation): a  $\delta = 0.05$  verifier accepts the false  $\forall x(x + 0 = 0)$ ,  $\forall x(Sx = x)$ ,  $\forall x\forall y(y = x)$  in 5 of 25 seeds, at  $n = 8$  or 16, never from  $n = 32$  (??). In pa §5.4 (one run, hand-picked candidates, two-part code)  $V_{\delta,0}$  accepts  $0 = S0$  at  $n = 10$  and 30 for every  $\delta \geq 3 \cdot 10^{-13}$ , because the lump  $F_0 + T_{\text{Ind}}$  leads (??). No contradiction with Theorem 1.11: for E2 it needs  $\delta_n \leq 2^{-226} R_{1/2}(n, 8) \delta' \asymp 2^{-226} n^{-3.5}$  ( $T^*$  costs 226.2 bits in the experiments’ code); in pa’s set  $\pi(Q + T_{\text{Ind}}) = 2^{-330}$ , and pa’s data are not drawn from a model generator.

## 1.6 The trichotomy: true, false, independent

Hänni’s working notes (Hänni, 2026) observe that a posterior over theories gives each sentence “a p(true/false/independent)” (`research/prior/hanni-solomonoff-axiom-induction.md`).

**Definition 1.15** (Belief, disbelief, independence). <sup>model §7; universal §1.6</sup> For a posterior  $\pi_D$  on a countable class and a sentence  $s$ , with  $\vdash$  derivability in  $K$  without a bound:

$$\begin{aligned} \text{Bel}(s) &:= \pi_D(T \vdash s), & \text{Dis}(s) &:= \pi_D(T \vdash \neg s), & \text{Indep}(s) &:= \pi_D(T \not\vdash s \text{ and } T \not\vdash \neg s), \\ \text{Inc} &:= \pi_D(T \text{ inconsistent}), & \text{Pl}(s) &:= 1 - \text{Dis}(s), & \text{Bel}_{\text{cons}}(s) &:= \pi_D(T \text{ consistent and } T \vdash s). \end{aligned}$$

The model track allows bounded derivability here; but  $\text{Th}_d(T)$  is not closed under consequence (a derivation of  $s$  has symbol size at least  $|s|$ ), so Proposition 1.16(a), (c), (d) and Proposition 1.17 need  $d = \infty$ . Proposition 1.18 holds for every  $d$ .

**Proposition 1.16** (Basic laws). [proved]<sup>model Prop 7.1</sup> (a)  $\text{Bel}(s) + \text{Dis}(s) + \text{Indep}(s) = 1 + \text{Inc}$ ; from now on the posterior is on consistent theories, so the three sum to 1. (b)  $\text{Dis}(s) = \text{Bel}(\neg s)$ ;  $\text{Bel}(\top) = 1$ ,  $\text{Bel}(\perp) = 0$ ;  $\text{Pl}(s) = \text{Bel}(s) + \text{Indep}(s)$ . (c) If  $s \vdash s'$  then  $\text{Bel}(s) \leq \text{Bel}(s')$ . (d) If  $\vdash \neg(s \wedge s')$  then  $\text{Bel}(s \vee s') \geq \text{Bel}(s) + \text{Bel}(s')$ .

**Proposition 1.17** (Bel is a belief function). [proved; framework known]<sup>model Prop 7.2</sup> For all sentences  $s_1, \dots, s_k$ ,  $\text{Bel}(s_1 \vee \dots \vee s_k) \geq \sum_{\emptyset \neq I \subseteq [k]} (-1)^{|I|+1} \text{Bel}(\bigwedge_{i \in I} s_i)$ . So Bel is a totally monotone capacity on the Lindenbaum algebra, with focal elements  $\text{Mod}(T)$ , and Pl is its plausibility (Shafer, 1976).

**Proposition 1.18** (Coherent bracketing). [proved]<sup>model Prop 7.3</sup> There is a probability  $P$  on complete consistent theories with  $\text{Bel}(s) \leq P(s) \leq \text{Pl}(s)$  for every  $s$ .

For one consistent  $T$ ,  $s \mapsto 1[T \vdash s]$  is multiplicative on conjunctions, which gives Proposition 1.16(c), (d) and Proposition 1.17; for Proposition 1.18, mix Lindenbaum completions of the theories with their posterior weights. Proofs of this subsection are in Appendix A.4.

**Example 1.19** (Renormalising is incoherent). [proved; computed]<sup>model Ex 7.4</sup> Renormalising means  $R(s) := \text{Bel}(s)/(\text{Bel}(s) + \text{Dis}(s))$ . With atoms  $a, b$  and theories  $\{a\}, \{b\}, \{\neg(a \wedge b)\}$  of posterior  $\frac{1}{3}$  each,  $R(a) = R(b) = 1$  but  $R(a \wedge b) = 0$ ; no probability does this.

**Example 1.20** (The 50/50 rule is incoherent). [proved; computed]<sup>model Ex 7.5</sup> Let  $H(s) := \text{Bel}(s) + \frac{1}{2}\text{Indep}(s)$ . Already for the empty theory over atoms  $b, c$ :  $H(b) = H(b \wedge c) = H(b \wedge \neg c) = \frac{1}{2}$ , while coherence needs  $P(b \wedge c) + P(b \wedge \neg c) = P(b)$ .

**Proposition 1.21** (When the 50/50 rule is coherent). [proved; computed]<sup>model Prop 7.6</sup> *Over finitely many propositional atoms,  $H = \text{Bel} + \frac{1}{2}\text{Indep}$  is coherent (equal to some probability on valuations, on every sentence) iff every theory of positive posterior mass has at most two models.*

A theory with one or two models contributes the uniform law on them; one with three or more models, two of which are  $v, v'$ , makes  $H(\{v, v'\}) < H(\{v\}) + H(\{v'\})$ . Linear programming agrees with the criterion in 3225/3225 posteriors on two atoms and 600/600 on three.

*Remark 1.22* (How the likelihood shapes Indep). [proved (from ???)]<sup>model §7</sup> Under Hänni’s 0/1 scores a consistent strengthening keeps its prior odds (??(b)), so Bel of a sentence  $\psi$  independent of the data stays at its prior level forever. Under L0 and L1, a theory that proves *and generates*  $\psi$  loses mass while  $\psi$  never appears: polynomially with Dirichlet weights (??), exponentially with fixed weights (??). So  $\text{Indep}(\psi)$  grows: with a generative likelihood, “the data never mention  $\psi$ ” is evidence against theories that would have mentioned it.

*Remark 1.23* (The  $\text{Bel}_{\text{cons}}$  verifier). [computed; transfer proved]<sup>universal §6.2, §11; editor, from model §4</sup> Inconsistent theories prove  $s$  and  $\neg s$  and inflate Bel and Dis alike: in universal’s c2 class at  $n = 0$ ,  $\text{Bel}(\forall x\varphi) \approx 0.69$  is almost all Inc; generative likelihoods remove such theories within a few data (??). Universal therefore recommends “accept  $s$  iff  $\text{Bel}_{\text{cons}}(s) \geq 1 - \delta$ ” (??). When this verifier accepts,  $V_{\delta,d}$  would accept at the same posterior, since  $\{T : T \not\models s\}$  is part of what it counts against  $s$ . The proofs of Theorems 1.2, 1.10 and 1.11 bound  $\pi_t(C_d^*)$  (or  $\pi_t(C_d^{\text{Dir}})$ ) from below for every sequence of truthful constraints, so all three transfer to it ( $\text{Bel}_{\text{cons}}$  read at level  $d$ ).

**Hänni’s three questions** (same file of his notes). (1) “i guess you could also naturally just get true/false by either just removing the independent fraction and renormalizing.” This is incoherent in general (Example 1.19), which confirms his “i guess we probably usually don’t get coherent probabilities either?”. (2) “instead of renormalizing, you could also turn the independent fraction into 50/50. tho i guess this has the downside of not giving coherent probabilities usually”. Confirmed: it fails already for one theory (Example 1.20), and it is coherent exactly when no theory of positive mass has more than two models (Proposition 1.21). (3) “hmm maybe this motivates not removing the independent fraction?” Yes.  $(\text{Bel}, \text{Pl})$  is a coherent lower and upper probability (Propositions 1.17 and 1.18). Like a semimeasure it keeps the mass on “undecided” instead of redistributing it; that is also the convention under which ?? holds with a constant. A coherent point value needs a distribution over the completions of each theory, which the trichotomy does not supply.

## A Soundness: proofs and details

Scripts named `ck` are the model track’s (`research/tracks/model/checks/`), `rk` its referee’s (`research/tracks/model/referee_code/`); the others are in `research/tracks/experiments/checks/`. Each is seeded and writes its output next to itself.

## A.1 Fixed weights

*Proof of Theorem 1.2.* This is the proof of IL Thm 4.15(b) (IL App. C.9), in the class form of IL Thm 10.9; the role of shadow-measurability there is played by the definition of  $C_d^*$ . Write  $L_t(T) := \prod_{i \leq n(t)} P_T(X_i) \prod_j c_j(T)$ .

*Step 1:  $C_d^*$  keeps the target's likelihood.* For  $T \in C_d^*$ ,  $P_T = P_{T^*}$ , and every constraint has  $c_j(T) = c_j(T^*) = 1$  because  $\text{Th}_d(T) = \text{Th}_d(T^*)$ . So  $L_t(T) = L_t(T^*)$ , which is positive almost surely. With  $Z_t := \sum_T \pi(T) L_t(T) / L_t(T^*)$  we get  $\pi_t(C_d^*) = W_d^* / Z_t$ .

*Step 2: constraints only help.*  $c_j(T^*) = 1$  and  $c_j(T) \leq 1$ , so pathwise  $Z_t \leq Z_{n(t)}^{\mathcal{H}}$ , where

$$Z_n^{\mathcal{H}} := \sum_T \pi(T) \prod_{i \leq n} \frac{P_T(X_i)}{P_{T^*}(X_i)} = \frac{M(X^n)}{P_{T^*}(X^n)}.$$

*Step 3: Ville.* Under  $P_{T^*}$ ,  $\mathbb{E}[P_T(X)/P_{T^*}(X)] = \sum_{x: P_{T^*}(x) > 0} P_T(x) \leq 1$ . By independence and Tonelli,  $Z^{\mathcal{H}}$  is a nonnegative supermartingale for  $\sigma(X_1, \dots, X_n)$  with  $Z_0^{\mathcal{H}} = 1$ . By Ville's inequality (IL Lemma C.4) the event  $G := \{\sup_n Z_n^{\mathcal{H}} < 1/\delta'\}$  has probability at least  $1 - \delta'$ . It depends only on the data.

*Step 4: conclusion.* On  $G$ , in every round and for every history of queries and truthful constraints,  $\pi_t(C_d^*) = W_d^* / Z_t \geq W_d^* / Z_{n(t)}^{\mathcal{H}} > W_d^* \delta' \geq \delta$ . If  $q \notin \text{Th}_d(T^*)$  then  $q \notin \text{Th}_d(T)$  for every  $T \in C_d^*$ , so  $\pi_t(\{T : q \notin \text{Th}_d(T)\}) > \delta$  and  $q$  is not accepted. The prover's knowledge and randomness enter only through the schedule and the constraints, so the conclusion covers a prover that knows the future data.  $\square$

**Computed (Theorem 1.2).** c4, Part 1: L0, fixed weights,  $\mathcal{Q}$  root law (.5, .3, .1, .1) on  $(0, S, +, \cdot)$ ,  $T^* = \{z+0 = z\}$  against  $\{z+0 = z, S0+0 = S0, 0+0 = S0\}$  (weights 0.49, 0.5, 0.01), which proves  $q := (0+0 = S0)$ ;  $\pi(T^*) = 0.3$ ; 20 000 runs of 300 data. The frequency of  $\{\exists n \leq 300 : \pi_n(T^*) \leq \delta\}$  was 0.044, 0.014, 0.004 against  $\delta/\pi(T^*) = 0.10, 0.033, 0.010$ . r7: nine hypotheses all proving the same invalid query,  $\pi(C^*) = 0.385$ ,  $n \leq 400$ : 0.0365, 0.0125, 0.0027 against  $\delta/\pi(C^*) = 0.130, 0.052, 0.013$ . The horizons are finite, so these check the bound, not its time-uniformity.

*Proof of Remark 1.3. IL's pair is not in the models.* Under L0 with full-support  $\mathcal{Q}$ ,  $\text{supp } P_T = \bigcup \text{inst}(T)$ , and  $\text{Th}_d$  depends only on the instance union; under L1 with parameters admissible  $\text{supp } P_T = \text{Th}(T)$  (??(c)). Either way  $\text{supp } P_{T'} \subseteq \text{supp } P_{T^*}$  gives  $\text{Th}_d(T') \subseteq \text{Th}_d(T^*)$ , so no  $T'$  generates only data of  $T^*$  and still proves a sentence  $T^*$  does not.

*The construction.* Let  $a, b, c$  be ground sentences with  $c \notin \text{Th}_d(T^*)$  and  $c \in \text{Th}_d(\{c\})$ , let  $T^* = \{a, b\}$  with fixed weights  $(1-u, u)$ ,  $T' = \{a, c\}$  with fixed weights  $(1-\varepsilon, \varepsilon)$ , prior  $(W^*, 1-W^*)$ , likelihood L0 and  $\delta = W^* \delta'$ . Then  $W_d^* = W^*$ . After  $t$  data equal to  $a$ ,  $r_t := P_{T'}(a^t) / P_{T^*}(a^t) = ((1-\varepsilon)/(1-u))^t$ , and  $c$  is accepted iff  $\pi_t(T^*) = W^* / (W^* + (1-W^*)r_t) \leq \delta$ , i.e. iff  $r_t \geq \theta := (1-W^* \delta') / ((1-W^*) \delta')$ . Let  $t^*$  be the least  $t$  with  $(1-u)^{-t} > \theta$  (IL Prop C.5). For  $t < t^*$ ,  $r_t < (1-u)^{-t} \leq \theta$ . At  $t^*$  the inequality  $(1-u)^{-t^*} > \theta$  is strict, so  $r_{t^*} > \theta$  for all  $\varepsilon$  below some  $\varepsilon_0 > 0$ . A datum  $b$  kills  $T'$ . So the prover that waits for  $t^*$  data and then asks  $c$  (an earlier query could be escalated, and the oracle's “no” would kill  $T'$ ) wins iff the first  $t^*$  data are  $a$ : probability  $(1-u)^{t^*}$ , exactly IL's value. By IL Prop C.5 it lies in  $[(1-u)/\theta, 1/\theta]$  with  $1/\theta = (1-W^*) \delta' / (1-W^* \delta')$ , so it tends to  $\delta'$  as  $u, W^* \rightarrow 0$ .  $\square$

**The computable verifier (Remark 1.4).** For a finite class and  $d < \infty$ ,  $\text{Th}_d(T)$  is decidable (finitely many derivations of symbol size  $\leq d$  up to renaming of parameters, axioms checked by matching), the likelihoods are computable reals (L0 by matching; L1 by ??(a)), and the

denominator of  $\pi_t$  is positive. So an interval of width  $\leq \varepsilon$  around the non-deriving mass can be computed; accept iff its upper end is  $\leq \delta$ . Acceptance then implies that the true mass is  $\leq \delta$  at the same posterior, and Step 4 applies verbatim. With truncated  $\mathcal{Q}$  the computation is exact.

**The refuted remarks (Remark 1.4).** (1) On a truncation  $\mathcal{H}_N \ni T^*$  the constant  $\pi(C_d^* \cap \mathcal{H}_N)/\pi(\mathcal{H}_N)$  is at least  $\pi(T^*)$  but can be below  $\pi(C_d^*)$ : with  $\mathcal{H} = \{T^*, T_2, T_3\}$ ,  $T_2 \in C_d^*$ , prior  $(0.01, 0.89, 0.10)$  and  $\mathcal{H}_N = \{T^*, T_3\}$  it is  $0.01/0.11 \approx 0.09$ , against 0.90. (2) L2 likelihoods are finite sums only when  $\mathcal{Q}$  has finite support: for  $T = \{P \rightarrow b\}$ ,  $P$  a 0-ary formula metavariable,  $P_T(b)$  at depth 1 sums  $\alpha_{\text{MP}} \mu(A) \mu(A \rightarrow b)$  over the infinitely many logical-axiom instances  $A$ .

**Proposition 1.5: computed.** c4, Part 2: the sound theories are ground equations for the 20 most likely human data (covering 0.956 of the human mass), with the human’s conditional probabilities as weights and prior 0.99 against  $T_{\text{esc}}$ . With  $\delta = 0.01$  and no escalation,  $S_0 = 0$  was accepted in 2000/2000 runs, median time 17 data; 0.341, 0.902, 0.986, 1.000 of runs by  $n = 10, 50, 100, 500$ . r3, Part 2: against a verifier that escalates every non-accepted query, a prover asking  $S_0 = 0$  in every round won 11/400 (only when the first datum was uncovered), the waiting prover 400/400. The track’s first version used the non-adaptive prover (referee m2).

## A.2 The cautious limit

*Proof of Proposition 1.6.* (a) The likelihood of  $T$  (fixed weights, or the Dirichlet marginal under L0) is positive iff every datum lies in  $\text{supp}_T$ , and  $\pi_n(T)$  is proportional to  $\pi(T)$  times it.  $V_{0,d}$  accepts  $q$  iff  $\pi_n(\{T : q \notin \text{Th}_d(T)\}) = 0$ , i.e. iff  $q \in \text{Th}_d(T)$  for every  $T$  of positive posterior mass. (b) Such a  $T^*$  is one of the theories in the intersection of (a), and truthful constraints keep it; so everything accepted lies in  $\text{Th}_d(T^*)$ . The accepted set of  $V_{\delta,d}$  grows with  $\delta$ . (c) The memoriser  $\text{Mem}(D_n)$ , made of ground templates for the distinct data, is charged by  $\pi$ , its support is the set of distinct data, and  $\text{Th}_d(\text{Mem}(D_n)) = \text{Th}_d(D_n)$ . Every  $T$  with  $D_n \subseteq \bigcup \text{inst}(T)$  has  $\text{Th}_d(D_n) \subseteq \text{Th}_d(T)$ , since  $\text{Th}_d$  is monotone in the axiom set. So the intersection in (a) is  $\text{Th}_d(D_n)$ ; this is AS Prop 6.4 ( $\text{Acc}_\infty(D) = D$ ) read on theorems. (d) A theory of positive posterior mass lies in  $H_k(\text{DT}^\circ)$  and covers  $D_n$ , so it is in the version space  $\text{VS}_k(D_n)$  of AS Def 2.6, and  $\bigcup \text{inst}(T) \supseteq \text{Acc}_k(D_n)$ . Monotonicity of  $\text{Th}_d$  gives the inclusion. *Strictness in a general class.* Take the class  $\{T_1, T_2\}$  with  $T_1 = \{c, a \wedge b\}$ ,  $T_2 = \{c, b \wedge a\}$  (ground), and  $D = \{c\}$ . Both theories prove  $a$ , so  $V_{0,d}$  accepts  $a$  for  $d$  large enough. The instance-level intersection is  $\{c\}$ , which does not prove  $a$ . Whether the inclusion can be strict for  $H_k(\text{DT}^\circ)$  itself is not settled (model §10, problem 10).  $\square$

**The two verifiers compared.** Table 1 contrasts the cautious verifier of AS with the thresholded verifier of Section 1 (model §5.5, with the weight convention stated).

## A.3 Dirichlet weights

*Proof of Remark 1.8.* This is the argument of model Thm 5.1(b''). Fix  $T$ . The map  $w \mapsto P_{T,w}$  is affine, so  $W_T := \{w : P_{T,w} = P_{T^*,w^*}\}$  is the intersection of the simplex with an affine subspace. If  $\text{Dir}_T(W_T) > 0$ , then  $W_T$  has positive Lebesgue measure in the simplex, so the affine map is constant on a set that spans the simplex’s affine hull, hence on the whole simplex, and every component of  $T$  has law  $P_{T^*,w^*}$ . (For  $|T| = 1$  the weight is trivial and the condition is  $\mathcal{Q}_\tau = P_{T^*,w^*}$ .) So the class has positive prior mass only if some template  $\sigma$  has  $\mathcal{Q}_\sigma = \sum_{\tau \in T^*} w_\tau^* \mathcal{Q}_\tau$ . For

|                                | cautious $k$ -union verifier (AS)                                                                      | Bayesian thresholded verifier                                                                                                         |
|--------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| assumptions for soundness      | realizability ( $R^* \in H_k$ ), bound $k \geq k'$                                                     | well-specified data law; $W_d^* \geq \delta/\delta'$ ; with Dirichlet weights, averaging over $w^*$ or a threshold shrinking with $n$ |
| soundness                      | deterministic; every prover, every time                                                                | probability $\geq 1 - \delta'$ ; every prover, every time (Theorems 1.2, 1.10 and 1.11)                                               |
| without the assumptions        | sound relative to $\text{cl}(R^*)$ (AS Lemma 2.8); without a bound accepts only the data (AS Prop 6.4) | can accept a false sentence with probability 1 (Proposition 1.5); at a fixed $w^*$ a constant threshold fails (Example 1.12)          |
| completeness                   | exact once an anchor is present (AS Lemma 2.8)                                                         | each theorem eventually, a.s. (Proposition 1.7; Dirichlet: a.e. $w^*$ ); no anchor, no bound; open under a shrinking threshold        |
| splits                         | sound splits survive any negatives (AS Thm 6.5)                                                        | exact ties at matched fixed weights; with learned weights Occam terms decide; usage statistics can reverse (?????)                    |
| depends on the target's length | no                                                                                                     | yes: $\delta \leq 2^{-\lambda \ell(T^*)} \delta'$ ; under Theorem 1.11 also on $ T^* $                                                |

Table 1: The cautious verifier of AS against the Bayesian thresholded verifier (model §5.5, with the weight convention stated).

each  $\sigma$ , linear independence of the component laws of  $T^*$  allows at most one such  $w^*$ , and there are countably many  $\sigma$ .  $\square$

*Proof of Lemma 1.9.* (a) Expand over labellings  $\ell : [n] \rightarrow T$  with counts  $c_\tau(\ell)$ :

$$P_{T,w}(x^n) = \sum_{\ell} \prod_{\tau} w_{\tau}^{c_{\tau}(\ell)} \prod_i q_{\ell_i}(x_i), \quad P_T^{\text{Dir}}(x^n) = \sum_{\ell} \mathbb{E}_{\text{Dir}} \left[ \prod_{\tau} w_{\tau}^{c_{\tau}(\ell)} \right] \prod_i q_{\ell_i}(x_i).$$

For each  $\ell$  the maximum of  $\prod_{\tau} v_{\tau}^{c_{\tau}}$  over the simplex is attained at  $v = c/n$ , so

$$\mathbb{E}_{\text{Dir}} \left[ \prod_{\tau} w_{\tau}^{c_{\tau}} \right] \geq R_{\alpha}(n, K) \prod_{\tau} (c_{\tau}/n)^{c_{\tau}} \geq R_{\alpha}(n, K) \prod_{\tau} w_{\tau}^{c_{\tau}}.$$

Sum over  $\ell$ . This termwise argument is experiments Prop X8(c).

(b) Fix  $c$ , put  $v := c/n$ ,  $\eta := 1/(n+1)$  and  $B := \{(1-\eta)v + \eta u : u \text{ in the simplex}\}$ . On  $B$ ,  $w_{\tau} \geq (1-\eta)v_{\tau}$ , so  $\prod_{\tau} w_{\tau}^{c_{\tau}} \geq (1-\eta)^n \prod_{\tau} v_{\tau}^{c_{\tau}} \geq e^{-1} \prod_{\tau} v_{\tau}^{c_{\tau}}$ , because  $(1-\eta)^n = (1+1/n)^{-n} \geq e^{-1}$ . The Dirichlet density  $\frac{\Gamma(A)}{\prod_{\tau} \Gamma(\alpha_{\tau})} \prod_{\tau} w_{\tau}^{\alpha_{\tau}-1}$  (with respect to Lebesgue measure on the first  $K-1$  coordinates) is at least  $\Gamma(A)/\prod_{\tau} \Gamma(\alpha_{\tau})$  on the simplex, since  $w_{\tau} \leq 1$  and  $\alpha_{\tau} \leq 1$ .  $B$  is the simplex scaled by  $\eta$ , so its volume is  $\eta^{K-1}/(K-1)!$ . Hence  $\mathbb{E}_{\text{Dir}}[\prod w^c] \geq e^{-1} \prod v^c \cdot \frac{\Gamma(A)}{\prod_{\tau} \Gamma(\alpha_{\tau})} \cdot \frac{\eta^{K-1}}{(K-1)!}$ ; divide by  $\prod v^c$  and minimise over  $c$ .

(d) Take  $c = (n, 0, \dots, 0)$ :  $\prod (c/n)^c = 1$  and  $\mathbb{E}_{\text{Dir}}[w_1^n] = \Gamma(K\alpha)\Gamma(\alpha+n)/(\Gamma(\alpha)\Gamma(K\alpha+n))$ ; then  $\Gamma(a+n)/\Gamma(b+n) \sim n^{a-b}$ .

(c) is the Krichevsky–Trofimov rate (Krichevsky and Trofimov, 1981); it is not re-proved here.  $\square$

**Computed (Lemma 1.9).** c11, Part A: exact Dirichlet marginals (a Bernstein-form dynamic programme, equal to quadrature to six decimals) for  $K = 2, 3$  with overlapping components on four letters,  $w^*$  interior and on the boundary,  $n \leq 3000$ . The largest value of regret minus bound

(b) was  $-2.81, -1.00, -6.45$  nats ( $K = 2, \alpha = \frac{1}{2}$ ;  $K = 2, \alpha = 1$ ;  $K = 3, \alpha = \frac{1}{2}$ ). Part A2: slopes of the exact  $\ln R_\alpha(n, K)$  against  $\ln n$  were  $-0.499, -0.993$  ( $K = 2, 3; \alpha = \frac{1}{2}$ ) and  $-0.993, -1.960$  ( $\alpha = 1$ ), against  $-(K - 1) \max(\frac{1}{2}, \alpha)$ . The experiments' `kt_regret`:  $-0.499, -0.998, -1.485$  for  $K = 2, 3, 4$  at  $\alpha = \frac{1}{2}$ .

*Proof of Theorem 1.10.* Steps 1, 2 and 4 of the proof of Theorem 1.2 go through with  $L_t(T) := P_T^{\text{Dir}}(X^{n(t)}) \prod_j c_j(T)$  and  $C_d^{\text{Dir}}$  in place of  $C_d^*$ . For Step 3 let  $M := \sum_T \pi(T) P_T^{\text{Dir}}$  and  $Z_n := M(X^n) / P_{T^*}^{\text{Dir}}(X^n)$ . Each  $P_T^{\text{Dir}}$  is a probability on sequences, consistent in  $n$ , so  $M$  is too, and under  $P_{T^*}^{\text{Dir}}$

$$\mathbb{E}[Z_{n+1} \mid X^n] = \sum_{x: P_{T^*}^{\text{Dir}}(X^n x) > 0} \frac{M(X^n x)}{P_{T^*}^{\text{Dir}}(X^n)} \leq \frac{M(X^n)}{P_{T^*}^{\text{Dir}}(X^n)} = Z_n.$$

Ville's inequality gives  $G := \{\sup_n Z_n < 1/\delta'\}$  with  $P_{T^*}^{\text{Dir}}(G) \geq 1 - \delta'$ . For the consequence,  $f(w^*) := P_{T^*, w^*}^\infty(G^c)$  has  $\int f d\text{Dir}_{T^*} = P_{T^*}^{\text{Dir}}(G^c) \leq \delta'$ , since  $P_{T^*}^{\text{Dir}}$  on infinite sequences is the mixture  $\int P_{T^*, w}^\infty \text{Dir}_{T^*}(dw)$ ; apply Markov's inequality.

*Pool version* (experiments Prop X8(a), (b); no escalation). Let  $F$  be the countable family,  $C := \sum_{T \in F} 2^{-\text{bits}(T)} \leq 1$  (Kraft for the experiments' code, ??),  $M_T := P_T^{\text{Dir}}$ , and  $Z_n := \sum_{T \in F} (2^{-\text{bits}(T)} / C) M_T(D_n) / M_{T^*}(D_n)$ , a nonnegative supermartingale under  $M_{T^*}$  with  $Z_0 = 1$  by the computation above. For every  $R_n \subseteq F$  with  $T^* \in R_n$ , however chosen, pathwise

$$\pi_R(T^* \mid D_n) = \frac{2^{-\text{bits}(T^*)} M_{T^*}(D_n)}{\sum_{T \in R_n} 2^{-\text{bits}(T)} M_T(D_n)} \geq \frac{2^{-\text{bits}(T^*)}}{C Z_n} \geq \frac{2^{-\text{bits}(T^*)}}{Z_n}.$$

Off the Ville event this exceeds  $2^{-\text{bits}(T^*)} \delta' \geq \delta$ . Accepting  $s$  with  $T^* \not\prec_J s$  needs mass  $\geq 1 - \delta$  on theories deriving  $s$ , hence  $\pi_R(T^* \mid D_n) \leq \delta$ . If  $F$  is a fixed pool  $H$  plus all memorisers,  $C \leq \sum_{T \in H} 2^{-\text{bits}(T)} + 1$ , because the memorisers' codes form a prefix-free set; use  $2^{-\text{bits}(T^*)} / C$  in place of  $2^{-\text{bits}(T^*)}$ .  $\square$

*Proof of Theorem 1.11.*  $Z'_n := M(X^n) / P_{T^*, w^*}(X^n)$  is a nonnegative supermartingale under the i.i.d. law  $P_{T^*, w^*}$  with  $Z'_0 = 1$ , by the computation above with  $P_{T^*, w^*}$  as reference law ( $M$  is a consistent probability on sequences). Let  $G := \{\sup_n Z'_n < 1/\delta'\}$ , so  $P(G) \geq 1 - \delta'$ . Members of  $C_d^{\text{Dir}}$  have  $T^*$ 's marginal likelihood and pass every constraint, and the other constraints are  $\leq 1$ , so

$$\pi_t(C_d^{\text{Dir}}) \geq \pi(C_d^{\text{Dir}}) \frac{P_{T^*}^{\text{Dir}}(X^n)}{M(X^n)} = \pi(C_d^{\text{Dir}}) \frac{P_{T^*}^{\text{Dir}}(X^n) / P_{T^*, w^*}(X^n)}{Z'_n} \geq \frac{\pi(C_d^{\text{Dir}}) R_\alpha(n, K)}{Z'_n},$$

with  $n = n(t)$ , by Lemma 1.9(a), which needs the likelihood to be linear in  $w$ . On  $G$  this exceeds  $\pi(C_d^{\text{Dir}}) R_\alpha(n, K) \delta' \geq \delta_n$ . No member of  $C_d^{\text{Dir}}$  derives an invalid  $q$ , so its non-deriving mass exceeds  $\delta_n$ .  $\square$

**Example 1.12: details.** *Why  $T^* \not\prec q$ .* In the term model in which  $+$  returns its first argument on  $(a, 0)$  exactly when  $a$  has root 0 or  $S$  (and returns the formal term  $a + b$  otherwise), every instance of  $0 + 0 = 0$  and  $(Sz) + 0 = Sz$  holds, but  $(0 \cdot 0) + 0$  denotes the term  $(0 \cdot 0) + 0 \neq 0 \cdot 0$ . *Why  $P_{T'} = (1 - \varepsilon) P_{T^*, w^*}$  on the data's support.* A datum is  $t + 0 = t$  with  $t = 0$  or  $t = St'$ . Under  $T'$  it has probability  $\mathcal{Q}(t)$ , i.e. 0.5 or  $(0.5 - \varepsilon) \mathcal{Q}(t')$ ; under  $P_{T^*, w^*}$  it has  $w_1^* = 0.5 / (1 - \varepsilon)$  or  $w_2^* \mathcal{Q}(t') = (0.5 - \varepsilon) \mathcal{Q}(t') / (1 - \varepsilon)$ . *The decomposition.*  $T^*$ 's instance sets are disjoint, so

$\ln P_{T^*, w^*}(D) - \ln P_{T^*, \hat{w}}(D) = \sum_f n_f \ln(w_f^* / \hat{w}_f) = -n \text{KL}(\hat{w} \| w^*)$ , with  $\hat{w}$  the empirical weights. With the previous point,

$$\ln \frac{P_{T'}(D)}{P_{T^*}^{\text{Dir}}(D)} = n \ln(1 - \varepsilon) + \left[ \ln P_{T^*, \hat{w}}(D) - \ln P_{T^*}^{\text{Dir}}(D) \right] - n \text{KL}(\hat{w} \| w^*).$$

The bracket is the regret of Lemma 1.9(c) at  $K = 2$ , also ??(b), and  $2n \text{KL}(\hat{w} \| w^*) \rightarrow \chi_1^2$  in law. *Not written out:* a version of that  $\chi^2$  limit uniform over the window  $1 \ll n \ll 1/\varepsilon$ , which the crossing argument needs. *The shrinking threshold* of the model track's run is  $\delta_n = \pi(T^*) \delta' \frac{1}{e\pi} (n+1)^{-1}$  with  $\delta' = 0.02$ , the bound of Lemma 1.9(b) at  $K = 2$ ,  $\alpha = \frac{1}{2}$  ( $1/(e\pi) = 0.1171$ ).

|                                                     | $\varepsilon$ | fixed $w^*$ , constant $\delta$ | $w^* \sim \text{Dir}(\frac{1}{2}, \frac{1}{2})$ , constant $\delta$ | fixed $w^*$ , shrinking $\delta$ |
|-----------------------------------------------------|---------------|---------------------------------|---------------------------------------------------------------------|----------------------------------|
| model (c11, every $n \leq 2 \cdot 10^6$ , 300 runs) | $10^{-5}$     | 0.997                           | 0.007                                                               | 0/300                            |
|                                                     | $10^{-6}$     | 1.000                           | 0.003                                                               | 0/300                            |
| experiments (check_fixed_weight, same)              | $10^{-5}$     | 0.977 (median first $n$ 8500)   | 0.003                                                               | 0/300                            |
|                                                     | $10^{-6}$     | 1.000 (median first $n$ 7525)   | 0.007                                                               | 0/300                            |
| model's referee (r3, geometric grid of $n$ )        | $10^{-5}$     | $\geq 0.980$                    | $\geq 0.008$                                                        | —                                |
|                                                     | $10^{-6}$     | $\geq 1.000$                    | $\geq 0.010$                                                        | —                                |

Table 2: Example 1.12: probability that the underivable  $q$  is ever accepted; nominal bound  $\delta/\pi(T^*) = 0.02$ . The referee checked a geometric grid of  $n$ , so its numbers are lower bounds. The differences between implementations are Monte Carlo and grid effects.

check\_fixed\_weight also matched its closed form against the posterior code (ln Bayes factors to  $4.1 \cdot 10^{-13}$ ) and checked Lemma 1.9(b) at  $K = 2$  for  $n \leq 2000$  (margin  $\geq 2.1$  nats).

## A.4 The trichotomy

Derivability is unbounded here (Definition 1.15).

*Proof of Proposition 1.16.* (a) A consistent theory proves at most one of  $s, \neg s$ , so it is counted once in  $\text{Bel}(s) + \text{Dis}(s) + \text{Indep}(s)$ ; an inconsistent one proves both and is counted twice. The sum is  $(1 - \text{Inc}) + 2\text{Inc}$ . (b)  $\text{Dis}(s) = \text{Bel}(\neg s)$  by definition; classically  $T \vdash \neg \neg s$  iff  $T \vdash s$ . On consistent theories  $\text{Bel}(\top) = 1$ ,  $\text{Bel}(\perp) = 0$ , and  $1 - \text{Dis} = \text{Bel} + \text{Indep}$  by (a). (c)  $\text{Th}(T)$  is closed under consequence. (d) A consistent  $T$  proves at most one of two incompatible sentences, and proves  $s \vee s'$  if it proves either.  $\square$

*Proof of Proposition 1.17.* For one consistent  $T$  put  $u(s) := 1[T \vdash s]$ . Closure under  $\wedge$ -introduction and elimination gives  $u(\bigwedge_{i \in I} s_i) = \prod_{i \in I} u(s_i)$ . So the right side equals  $\sum_{\emptyset \neq I} (-1)^{|I|+1} \prod_{i \in I} u(s_i) = 1 - \prod_i (1 - u(s_i)) = 1[T \text{ proves some } s_i] \leq u(s_1 \vee \dots \vee s_k)$ . The inequality is linear in  $u$ , so it survives the mixture  $\text{Bel} = \sum_T \pi_D(T) u_T$ . By completeness,  $u_T(s) = 1[\text{Mod}(T) \subseteq \text{Mod}(s)]$ , which identifies the focal elements.  $\square$

*Proof of Proposition 1.18.* For each  $T$  in the posterior's (countable) support choose a complete consistent extension  $\bar{T}$  (Lindenbaum), and put  $P(s) := \sum_T \pi_D(T) 1[s \in \bar{T}]$ , a mixture of point masses on complete theories. Then  $1[T \vdash s] \leq 1[s \in \bar{T}] \leq 1 - 1[T \vdash \neg s]$ , the second because  $\bar{T}$  is consistent and contains  $\neg s$  when  $T \vdash \neg s$ . Average over  $T$ . The argument uses only  $\text{Th}_d(T) \subseteq \text{Th}(T) \subseteq \bar{T}$ , so it holds for bounded derivability as well.  $\square$

*Proof of Examples 1.19 and 1.20.* Example 1.19:  $\text{Bel}(a) = \frac{1}{3}$  (from  $\{a\}$ ) and  $\text{Dis}(a) = 0$ , since no theory proves  $\neg a$ ; so  $R(a) = 1$ , and likewise  $R(b) = 1$ . No theory proves  $a \wedge b$ , and  $\{\neg(a \wedge b)\}$  refutes it, so  $R(a \wedge b) = 0$ . A probability with  $P(a) = P(b) = 1$  has  $P(a \wedge b) = 1$ . Example 1.20: the empty theory proves none of  $b$ ,  $b \wedge c$ ,  $b \wedge \neg c$  or their negations, so each gets  $H = \frac{1}{2}$ , and  $\frac{1}{2} + \frac{1}{2} \neq \frac{1}{2}$ .  $\square$

*Proof of Proposition 1.21.* Identify a consistent theory with its nonempty set of models  $\text{Mod}(T)$  among the valuations, and a sentence with its set of models; over finitely many atoms every set of valuations is a sentence. For one theory

$$H_T(s) = 1[\text{Mod}(T) \subseteq s] + \frac{1}{2} 1[\text{Mod}(T) \text{ meets both } s \text{ and its complement}],$$

and  $H = \sum_T \pi_D(T) H_T$ . *If.* With  $|\text{Mod}(T)| = 1$ ,  $H_T$  is the point mass on the model. With  $\text{Mod}(T) = \{v, v'\}$ ,  $H_T(s)$  is 1,  $\frac{1}{2}$  or 0 as  $s$  contains both, one or none of  $v, v'$ , i.e.  $H_T = \frac{1}{2}(\delta_v + \delta_{v'})$ . A mixture of probabilities is a probability. *Only if.* For valuations  $v \neq v'$  put  $D_T := H_T(\{v, v'\}) - H_T(\{v\}) - H_T(\{v'\})$ . If  $v, v' \in \text{Mod}(T)$  and  $|\text{Mod}(T)| \geq 3$ , all three sets meet  $\text{Mod}(T)$  and its complement, so  $D_T = \frac{1}{2} - \frac{1}{2} - \frac{1}{2} = -\frac{1}{2}$ . In every other case  $D_T = 0$ : if  $\text{Mod}(T) = \{v, v'\}$ ,  $D_T = 1 - \frac{1}{2} - \frac{1}{2}$ ; if exactly one of them, say  $v$ , is a model, then  $H_T(\{v'\}) = 0$  and  $H_T(\{v, v'\}) = H_T(\{v\})$ , because  $\{v, v'\}$  and  $\{v\}$  meet  $\text{Mod}(T)$  and its complement in the same way; if neither is a model, all three terms are 0. So if some  $T_0$  with  $\pi_D(T_0) > 0$  has three or more models and  $v, v'$  are two of them,  $\sum_T \pi_D(T) D_T \leq -\frac{1}{2} \pi_D(T_0) < 0$ , whereas a probability gives 0.  $\square$

**Linear-programming checks.** **c5:** Examples 1.19 and 1.20 are infeasible as probabilities; 300 random posteriors on three atoms gave no violation of 2- or 3-monotonicity and no bracketing failure; renormalisation was incoherent in 90/300, the 50/50 rule in 290/300. **r5** (independent generator): no violation of 2-, 3- or 4-monotonicity on two atoms, no bracketing failure with other completions; the 50/50 rule was coherent in 12/300 trials, exactly those in which every theory had at most two models. **c14:** the criterion of Proposition 1.21 agrees with LP coherence in 3225/3225 posteriors on two atoms (all one- and two-theory supports plus 3000 random) and 600/600 on three. (Model §7 also credits c5 with 4-monotonicity; c5's output records only 2 and 3, and 4 was checked by the referee on two atoms.)

*Remark A.1* (Renormalisation and complete theories). [refuted; computed]<sup>model §7 (referee m4)</sup> The model track's first version said: "the coherent cases [of renormalisation] are essentially those where every theory is complete". **Refuted:** in the referee's trials renormalisation was coherent in 199 of 300 posteriors, only 2 of them with all theories complete; in **c14** it was coherent in 1761 cases on two atoms (218 with all theories complete) and 306 on three atoms (42). For the 50/50 rule the exact condition is Proposition 1.21; for renormalisation no characterisation is given.

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